# The Computational Investigation of Fuel Molecular Structures via Python and Avogadro

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## Introduction

As populations rise and countries around the world become increasingly industrialized, global energy needs are growing rapidly. In contemporary society, most of the world's energy is taken from fossil fuels such as petroleum and natural gas. However, a major drawback to this method is that burning such fuels creates and releases carbon dioxide, a gas that traps heat in Earth's atmosphere; this is a large contributing factor in global warming and climate change. The main culprit of these emissions is sources of transportation such as cars, trucks, ships, and planes, which all run on petroleum-based fuels (Umenweke and Santillan-Jimenez).

In order to reduce emissions and the use of fossil fuels, numerous scientists and engineers have explored and studied alternative sources of fuel made from renewable biomass. These are called biofuels, which can be naturally produced from biodegradable materials such as corn, sugarcane, leftover crop materials, cooking oil, and other sources. The most common types of biofuels include ethanol and butanol, which are able to be mixed with gasoline or used as fuel on their own. When compared to harmful fossil fuels, biofuels can significantly lower greenhouse gas emissions, making energy use more eco-friendly ("Biofuels Basics"). In fact, according to the U.S. Energy Information Administration (EIA), new rules have been implemented to require more biofuel blending, and are predicted to push ethanol, biodiesel, and renewable diesel production to record highs in 2026 and beyond. The EIA speculates that fuel ethanol production will continue to increase, resulting in renewable diesel production growing over 20% (Troderman). Because more governments and companies are beginning to search for cleaner energy options, it is imperative to understand how biofuel molecules actually function.

The performance of a fuel is dependent on its molecular structure, specifically: size, molecular weight, polarity, and forces between molecules. These characteristics all affect significant properties such as boiling point, how the fuel needs to be stored, how it is transported, and how well it works as a fuel. Chemical engineers base their studies on these factors, attempting to create improved fuel production systems, separation processes like distillation, and more sustainable energy technologies. The use of software, namely Avogadro, allows scientists and engineers to view, build, and render three-dimensional models of molecules, accelerating research into how differences in structure influence molecular behavior.

Through comparing hydrocarbon fuels such as methane, ethane, propane, and butane with biofuels like ethanol and butanol, this project will explore how molecular structure affects fuel properties and their significance in chemical engineering.

## Purpose

The purpose of this project is to explore how molecular structure affects the physical properties of typical hydrocarbon fuels and biofuels using Avogadro and data analysis. Through comparison between the molecules, such as their size and functional groups, this project will demonstrate how these factors affect how certain fuels behave and how they are incorporated in chemical engineering.

## Research Question

How does molecular structure affect the physical properties of hydrocarbon fuels and biofuels, and how can this be applied to fuel storage, transportation, and separation processes?

## Hypothesis

As the length of the carbon chain increases, the boiling point will increase since bigger molecules undergo stronger intermolecular forces. As well as this, alcohols comprising oxygen atoms will demonstrate increased boiling points than similar hydrocarbons due to hydrogen bonding.

## Methods

The six fuels: methane ($CH_4$), ethane ($C_2H_6$), propane ($C_3H_8$), butane ($C_4H_{10}$), ethanol ($C_2H_5OH$), and butanol ($C_4H_9OH$) were used and analyzed throughout the experiment. These molecules were specifically chosen because they are common hydrocarbon fuels and biofuels used in transportation and energy.

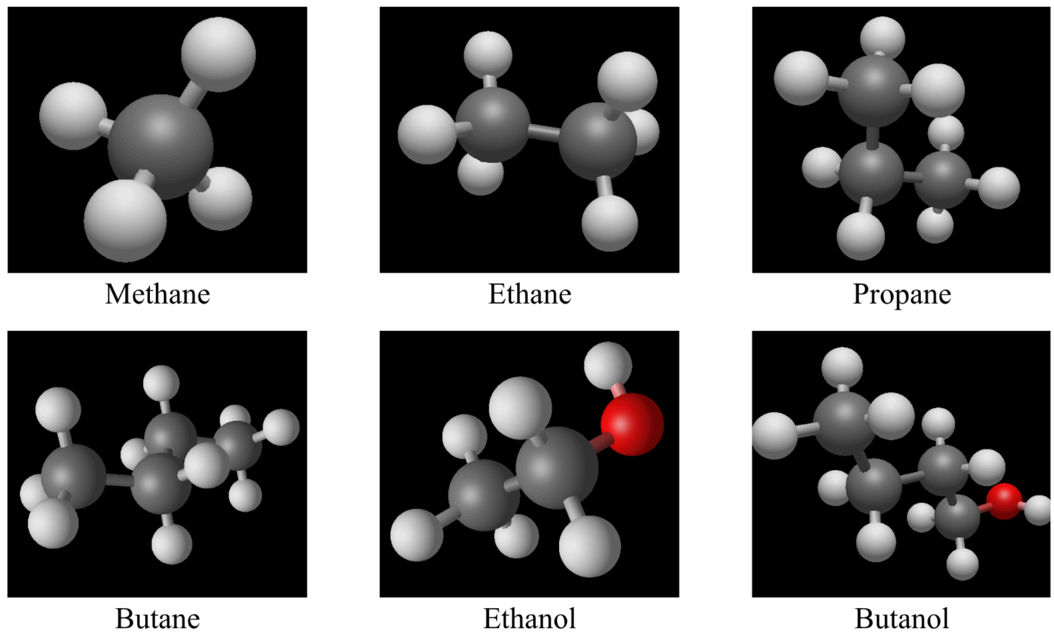


**Figure 1.** Optimized molecular models of methane, ethane, propane, butane, ethanol, and butanol created using Avogadro

The molecular structures were created from Avogadro's modeling tools, where each molecule's geometry was optimized to generate a stable three-dimensional structure. Screenshots of each molecule were taken for comparison.

The bond lengths and angles were measured using Avogadro's measurement tools, where carbon-carbon (C-C), carbon-hydrogen (C-H), carbon-oxygen (C-O), and oxygen-hydrogen (O-H) bonds were evaluated. Bond angles such as H-C-H, C-C-C, and C-C-O were also measured.

The physical properties, such as molecular weight and boiling point, were acquired from the respective Boiling Point and Molecular Weight Calculators. The data were organized into tables and analyzed using Claude AI's Python coding capabilities. While minor errors arose while formulating the graphs using AI-assisted code generation, they were quickly resolved through practical understanding of the Pandas library for data organization and Matplotlib for data visualization in Python. Graphs were then generated in Google Colab to visualize relationships among carbon chain length, molecular weight, oxygen content, and boiling point. The complete source code and workflow have been uploaded to GitHub to allow reproduction and exploration of the results of this project. The repository can be accessed at: https://github.com/matthewdengg/computational-fuel-analysis

## Results

The six fuels were generated and analyzed in Avogadro, specifically the four hydrocarbon fuels and two biofuels.

| Molecule | Formula | Contains Oxygen? | Functional Group |
|---|---|---|---|
| Methane | $CH_4$ | No | Alkane |
| Ethane | $C_2H_6$ | No | Alkane |
| Propane | $C_3H_8$ | No | Alkane |
| Butane | $C_4H_{10}$ | No | Alkane |
| Ethanol | $C_2H_5OH$ | Yes | Alcohol (-OH) |
| Butonal | $C_4H_9OH$ | Yes | Alcohol (-OH) |

**Table 1.** Molecular properties and characteristics of the analyzed fuel molecules

Following the examination, methane, ethane, propane, and butane were found to be all alkanes made up of only carbon and hydrogen atoms. Ethanol and butanol, on the other hand, contained oxygen in the form of a hydroxyl (-OH) functional group.

The bond length was generally consistent for all molecules.

| Molecule | Measured Bond | Length (Å) |
|---|---|---|
| Methane | C-H | 1.109 |
| Ethane | C-C, C-H | 1.517, 1.110 |
| Propane | C-C | 1.52 |
| Butane | C-C | 1.52 |
| Ethanol | C-C, C-O, O-H | 1.519, 1.110, 1.398 |
| Butonal | C-C, C-O, O-H | 1.524, 1.111, 1.398 |

**Table 2**. Bond length measurements across the analyzed molecules

C-C bonds ranged from 1.517 Å to 1.524 Å, while C-H bonds ranged from 1.109 Å to 1.111 Å. On top of this, the C-O and O-H bonds remained approximately 1.398 Å in both alcohol molecules. These findings demonstrate that the bond lengths between each atom had little to no variation despite differences in molecular size and composition.

Likewise, the bond angles showed little difference.

| Molecule | Measured Angle | Bond Angle (°) |
|---|---|---|
| Methane | H-C-H | 109.471 |
| Ethane | H-C-H | 109.119 |
| Propane | C-C-C | 111.324 |
| Ethanol | C-C-O | 110.499 |
| Butonal | C-C-O | 110.327 |

**Table 3.** Bond angle measurements across the analyzed molecules

Methane had an H-C-H bond angle of 109.471°, while ethane measured 109.119°. Propane had a C-C-C bond angle of 111.324°, and the alcohols exhibited C-C-O bond angles of 110.499° for ethanol and 110.327° for butanol. As a whole, the bond angles were extremely close to the ideal tetrahedral angle of 109.5°, revealing that the molecular geometry was similar between each molecule.

However, the physical property data displayed a much more noticeable variation than the geometric measurements.

| Molecule | Weight (g/mol) | Boiling Point (°C) |
|---|---|---|
| Methane | 16.04 | -161.5 |
| Ethane | 30.07 | -88.6 |
| Propane | 44.1 | -42.1 |
| Butane | 58.12 | -0.5 |
| Ethanol | 46.07 | 78.4 |
| Butonal | 74.12 | 117.1 |

**Table 4.** Physical properties of the analyzed fuel molecules

The molecular weight dramatically increased from methane (16.04 g/mol) to butanol (74.12 g/mol). In addition, the boiling point increased with the size of each molecule. Boiling points rose from -161.5°C for methane to -0.5°C for butane, with the alcohols having significantly higher boiling points than hydrocarbons with similar carbon chain lengths.

Specifically, ethanol had a boiling point of 78.4°C compared to -88.6°C for ethane, and butanol had a boiling point of 117.1°C, whereas butane only had a boiling point of -0.5°C.

The first graph depicted a positive relationship between carbon chain length and boiling point.

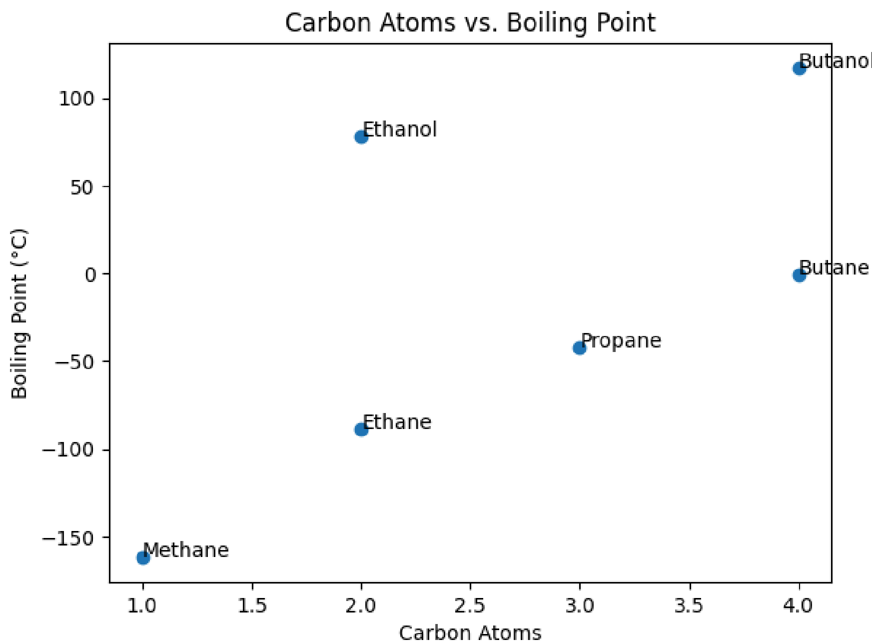


**Figure 2.** Relationship between carbon atoms and boiling point

As shown in Figure 2, boiling point increased as carbon atoms increased.

A similar relationship was found between molecular weight and boiling point.

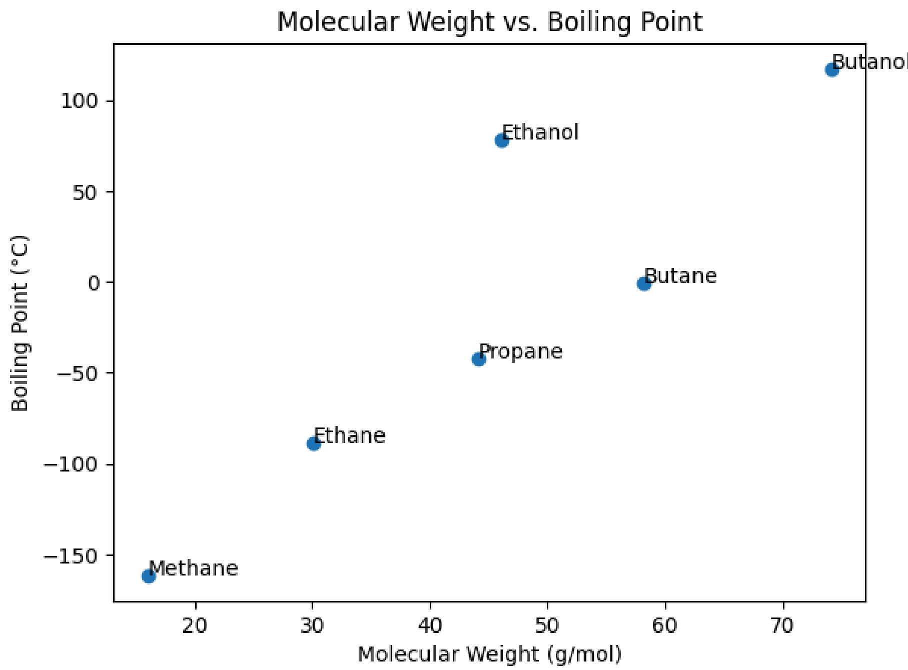


**Figure 3.** Relationship between molecular weight and boiling point

Figure 3 clearly shows that molecules with higher molecular weights most often had higher boiling points.

In the end, the molecules that contained oxygen consistently demonstrated considerably higher boiling points than hydrocarbons of similar size.

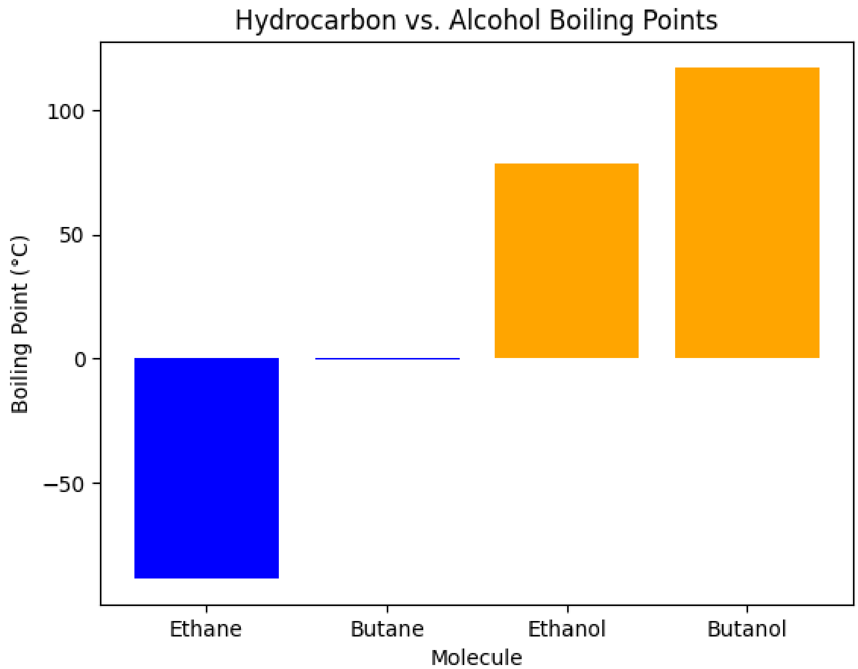


**Figure 4.** Comparison between hydrocarbon and alcohol boiling points

## Discussion

The results support the hypothesis, since as the carbon chain length increased from methane to butane, the molecular weight and boiling point increased too. Moreover, as the carbon chain length increased from methane to butane, the molecular weight and boiling point increased accordingly. As such, as intermolecular attractions increase in strength, more energy is needed to separate the molecules during boiling, resulting in a consistently higher boiling point.

As shown in Tables 2 and 3, the bond lengths and angles remained relatively constant despite different molecules being measured. C-C bond lengths stayed around 1.52 Å, C-H bonds averaged close to 1.11 Å, and most bond angles were close to the ideal tetrahedral angle of 109.5°. This displays that molecular geometry virtually had no difference between the simple hydrocarbon and alcohol molecules, meaning that the differences in boiling point are not a product of bond lengths or angles alone.

With that being said, the presence of oxygen within the molecules has demonstrated a much more profound impact on fuel behavior. Ethanol and butanol, which both contain hydroxyl functional groups, can form hydrogen bonds between molecules. Hydrogen bonding is significantly stronger than the London dispersion forces commonly found in hydrocarbons. Consequently, ethanol and butanol exhibited substantially higher boiling points than ethane and butane despite having similar carbon chain lengths. This displays how intermolecular forces are a much more prevalent factor in determining boiling point than small variations in molecular geometry.

This knowledge is crucial for chemical engineers, as they must understand how molecular structures influence physical properties when designing fuel storage systems, transportation methods, and industrial separation processes such as distillation. For example, butane's low boiling point means that it must be stored under moderate pressure in sealed canisters, the most common method being in portable camping stoves and household lighters. In contrast, butanol's high boiling point makes it much easier to handle and distribute as a biofuel. Similarly, the drastic difference in boiling points between ethanol and butane allows them to be separated efficiently using fractional distillation, a physical process used in fuel refineries to separate a complex mixture of liquids into individual components based on their boiling points (Darcel).

Overall, the inquiry demonstrates that though molecular geometry gives a relative basis in understanding atomic arrangement, it is ultimately molecular composition and intermolecular forces that drive the physical properties of fuels and determine their applications in the real-world.

## Conclusion

This study evaluated how molecular structure affected the physical properties of common hydrocarbon fuels and biofuels using Avogadro molecular modeling and Python data analysis. The results directly supported the hypothesis that increasing carbon chain lengths results in higher boiling points since larger molecules encounter stronger intermolecular forces. Moreover, alcohols containing hydroxyl groups had significantly higher boiling points than hydrocarbons with similar carbon chain lengths due to hydrogen bonding.

Despite bond lengths and bond angles being relatively similar among the molecules, molecular composition had a much stronger impact on physical properties. The presence of an oxygen atom leads to much stronger intermolecular attractions, which dramatically increased boiling point. This indicates that intermolecular forces are the main factor affecting fuel behavior.

This project highlights the power of computational tools such as Avogadro and Python for visualizing molecular structures and exploring the relationship between molecular design and physical properties. Understanding these connections is vital for chemical engineers working to improve fuel storage, transportation, separation processes, and the development of increasingly sustainable energy technologies.

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